Infor Mongoose
System Administration Guide

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About This Guide

This guide contains background or supplemental information to answer questions you may encounter as you manage and maintain Infor Mongoose.

This manual is intended as a reference. For instructions on how to install the Mongoose system, see the *Infor Mongoose Installation Guide*. For information about using the Mongoose system, see the Mongoose online help. Consult this manual when you have questions about Mongoose design or architecture.

Additional Infor Documentation

The most current version of all documentation is available on the Infor support web pages (see “Contacting Infor Support" below).

Online Help

Mongoose online help gives you instant access to procedures and information about forms and fields. You can access Help from Mongoose screens; just click the Help button.

Contacting Infor

If you have questions about Infor products, go to the Infor Xtreme Support portal at www.infor.com/inforxtreme.

If we update this document after the product release, we will post the new version on this Web site. We recommend that you check this Web site periodically for updated documentation.

If you have comments about Infor documentation, contact documentation@infor.com.

Scheduling an Installation

To schedule a Mongoose installation, contact your customer sales representative or district service manager.

Planning Your Communication

To make sure the correct analyst is assigned to your case and to expedite the resolution of your questions, please have the following information available when you call us:

- Your company name and phone number
- Mongoose version
- Database software version and release, if applicable
- Platform or environment
- What you were doing
- If you are calling back on an existing case, the case number

Signing Up for Support

If you are not currently on support and would like more information on your support options, please call your customer account representative. If you are not sure who your account representative is, contact Infor Customer Service.

Chapter 1: Overview

1

Infor Mongoose

Infor Mongoose is a comprehensive relational database application development framework for building and deploying Infor-ready (Workspace & ION) enterprise applications. The central goal of Mongoose is to minimize the amount of program coding required by developers.

Mongoose:

- Is an Infor layer built with, and directly onto, native Microsoft standard technologies. It is written entirely in .NET "managed code", and the database and application server/web server layers are implemented natively in core Microsoft server technologies such as SQL Server and IIS.
- Streamlines the process of creating new code by enabling you to use familiar and standard Microsoft .NET technologies and tools.
- Uses a model defined by metadata (data defining the application behavior, stored in rows and columns in databases); the user interface (forms); the Smart Business Layer, which consists of the mid-tier business objects (IDOs,or Intelligent Data Objects); and business processes (Application Event Handlers). In the metadata approach, changes can be kept separate from the base application and your changes automatically upgrade using tools like FormSync.
- Is fully cloud-enabled, so you have many future deployment options as your business grows and changes.

Chapter 2: Architecture

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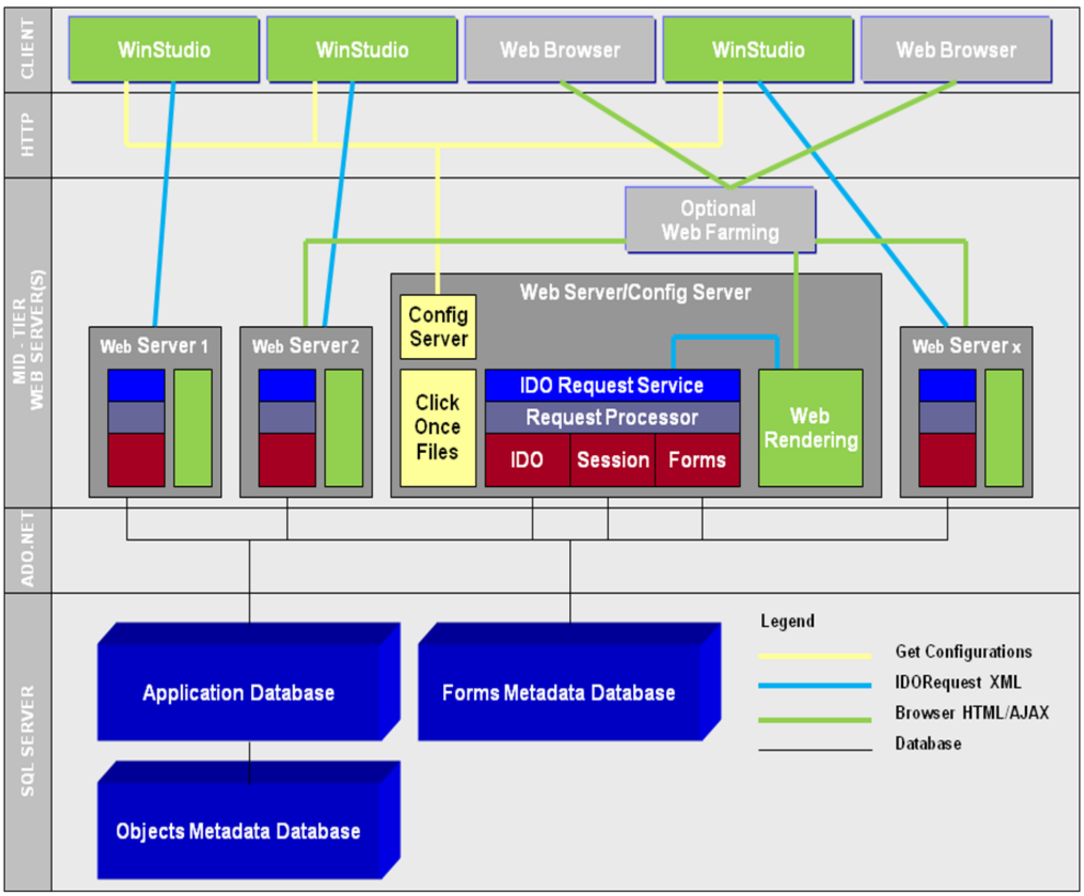
Infor Mongoose Architecture

Below is a graphical representation of the Mongoose architecture.



SQL Server

Microsoft SQL Server is responsible for things such as:

- Maintaining the relationships between data in the database
- Ensuring that data is stored correctly and that the rules defining data relationships are not violated
- Managing data and user security
- Optimizing server function performance
- Recovering all data to a point of known consistency in case of system failures.

SQL Server is primarily responsible for managing databases that interact with enterprise business systems. Some of the components that make up these databases are:

- Tables
- Views
- Stored Procedures
- Triggers
- Constraints
- Indexes
- Keys
- User Defined Data Types.

Application Database

The application database is one of the three main databases (the forms database and the objects database are the others) that comprise the “back-end” of Mongoose. The application database holds all of the application data for Mongoose. It is made up of of tables, stored procedures, triggers, user defined data types, and indexes. This information is stored on the database server.

Forms Database

The forms database holds all of the information about each form in Mongoose. This information is stored on the database server.

For example, components such as text boxes, labels, tabs, radio groups, buttons, and all of the associated properties (such as size, color, validation, events) are kept in the forms database in a multitude of tables.

Objects Database

The objects database stores the IDO metadata.

Clients

The client is where the user interacts with the application data and can customize forms.

End-User, Smart Client or Admin Client

Clients connect to the utility server. You can have a client reside on its own machine, or you can use the admin client that resides on the utility server. For the Smart Client (or the Smart Client via web browser), users get the URLs from the Mongoose Administrator. The URLs are set up on the utility server when Mongoose is installed.

Web Client

The web client renders forms in browsers and can run on multiple platforms and devices. It does not support form editing (design mode).

Intelligent Data Objects (IDOs)

Intelligent Data Objects present data from the application database or other sources as business objects that may be consumed by WinStudio forms or external applications that integrate with a Mongoose application. These objects are defined in metadata and are responsible for querying, creating, updating, and deleting data in the application database. IDOs may also publish methods that execute .NET code in the IDO runtime or stored procedures in the database. A developer may extend or override standard IDO functionality by associating an IDO with an extension class. For more information about extension classes, see the online help.

Chapter 3: Infor Framework TaskMan and Background Tasks

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This chapter describes the Infor Framework TaskMan service: how it works, how to set it up after installation, how to control tasks and reports, and how to troubleshoot problems with task management, report previews, or printing.

Overview

TaskMan (which is different from the Windows Task Manager) is a Windows service that:

- Polls the application database
- Executes SQL stored procedures
- Launches applications
- Processes and generates reports (for more information about TaskMan specific to reports, see the *Creating and Customizing Reports* guide)
- Runs IDO methods

Polling the Application Database

TaskMan polls the application database to identify new tasks to run. It uses the polling interval defined on the **Intranets** form. On startup it connects to the application database.

When a task is found that is waiting to be run, the task information (including executable, type of task parameters, and user name) is queried and the appropriate processing is performed.

Executing SQL Stored Procedures

For a SQL stored procedure, the TaskMan launches a database process thread. The database process then connects to the application database. It logs in as the Mongoose user who submitted the task. The stored procedure is then executed using the process connection.

Launching Applications

TaskMan launches a system process to execute any valid Windows application such as EDI (EDIImporter.EXE) and the Mongoose Scheduler (AIM_BATS.EXE), enabling the application to carry out its processing. The application is executed under the user account used by the TaskMan service.

Setup During Installation

TaskMan is installed and configured as part of the Mongoose installation. We recommend that you install TaskMan on the utility server. Much of the TaskMan setup is done behind the scenes. For more information, see the *Product Name (variable) Installation Guide*.

The installation process:

- Installs TaskMan on the utility server, which places the RunReport.exe program, the TaskMan.exe program, and other necessary files in the correct installation folder.
- Sets up the following subfolders within the ***installationFolder*\Report** folder:
    - **Errors** - Errors generated during report processing are written to this directory, under subfolders that match the IDs of the users who submitted the reports (for example, \Report\Errors\johsmi).
    - **OutputFiles** - The finished report output files are written to this folder, under subfolders that match the IDs of the users who submitted the reports. File extensions such as .DOC or .HTM indicate the format of the report.
    - **ParmFiles** - This folder contains XML files containing report parameter information. These files are written to this directory, under subfolders that match the IDs of the users who submitted the reports (for example, Report\ParmFiles\johsmi).
    - **Reports** - All report definitions (.rdl files) are placed here.

  Error and output file names include the site name and task number, using the format *taskname_site_tasknumber*. If TaskMan is configured to run with two or more databases that have the same site specification (for example, *test* and *production* databases) and name clashes occur, a single digit is appended to the file name to eliminate the clash. For example: DatabaseSchemaReport_OH_150844_2.

- Creates TaskMan as a Windows service on the utility server, set for automatic startup. If you enter a service logon ID and password during the TaskMan configuration part of the installation, that domain user ID and password are used to determine the Windows user account that controls the service.

  After installation is complete, you can also set the service logon ID and password from the **Services** dialog box. See your operating system help for how to do that.

- Determines which Strings table in the Forms database to use when displaying report form labels. This is a language-dependent setting that appears in the **Sites** form.

The Service Configuation Manager utility also allows you to configure additional OLE databases for monitoring by TaskMan. For Windows Server 2008, select **Start > Programs > Infor > Setup > Server Service Configuration**. For Windows Server 2012, the Server Service Configuration is found under Apps. Online help is available.

Setup After Installation

Before users can print or preview reports, or run other tasks as background processes, you need to perform the following setup tasks.

- Configure a printer on the utility server
- Define **Report/TaskMan** settings on the **Intranets** form
- Define e-mail addresses if required
- Define report options
- Define any excluded tasks

Configuring Printers

1. Log in to the server where TaskMan is installed (usually the utility server) using the same system administrator account that you used to install and run the Infor Framework TaskMan service.
2. Configure a default printer for the server, plus any other printers that you want TaskMan to print to. Printers must be configured using their UNC name (*\\PrintServer\PrinterName*).

Defining **Report/TaskMan** Settings on the **Intranets** Form

The **Intranets** form has a **Reports/Taskman** tab whose fields are used to set default values for TaskMan and for displaying or printing reports. You can leave most of these fields blank and accept the default values, but you *must* enter a TaskMan path for report previews. Some of these values can also be overridden on other forms. For more information, see the online help.

- **TaskMan Path** - The path to the folder containing the TaskMan utility.
- **Report URL** - The path to the folder or internet area where output files should be placed.
- **Polling Interval** - The interval at which TaskMan queries the databases to see if tasks are waiting to run.
- **Maximum Concurrent Tasks** - The maximum number of tasks that TaskMan can run simultaneously.
- **Maximum Concurrent Report Tasks** - The maximum number of report tasks that TaskMan can process simultaneously.
- **Send E-mail Notification** - Enables e-mail notification for all users when reports are printed. For additional requirements, see “Setting up the Utility Server to Send E-mail Notifications."
- **Output and Preview formats** - The default format used when creating reports or previews.

For information about the other fields and options on this tab, see the online help for the **Intranets** form.

Setting up the Utility Server to Send E-mail Notifications

Configure the system by entering information into these fields on the **Intranets** form:

- **Server**
- **Server Port**
- **From Email**
- **Authentication**
- **User Name**
- **Password**
- **Delivery Method**
- **Enable SSL**
- **Pickup Directory**

Enabling E-mail Attachments

To enable e-mail attachments:

1. Open the **Intranets** form.
2. Select the desired site.
3. On the **Reports/TaskMan** tab, make sure that the **Send Email Notification** field is selected.

Activating E-mail Notifications

For each report designed to send e-mail notifications, perform these tasks:

1. Open the **Report Options** form.
2. For each report (Task Name) that will send e-mail notifications, select **Yes** in the **E-mail Notification** field.

Note: Depending on how you have organized your report options, you might have to create new records for the reports that will use e-mail notifications.

Entering User E-mail Addresses

To enter e-mail addresses for users:

1. Open the **Users** form.
2. For each user who is to receive e-mail report notification, ensure that a valid e-mail address has been entered.

Defining Report Options

You can set these options for specific users and specific reports in the **Report Options** form:

- **User** - The user ID (if any) to which these report options apply.
- **Task Name** - The name of the task (if any) to which these options apply.
- **Output Format** - The format to use for report outputs for this user/task.

  This option overrides the format entered in the **Intranets** form.
- **Printer Name** - The name of the printer to use when the designated user submits the specified report. This printer must be defined as a network printer and must be configured on the server where TaskMan is installed. Also, the user ID set up as the owner of the TaskMan service must have print privileges for each printer defined here.
- **E-mail Notification** - Indicates whether an e-mail notification is sent to this user when this report is processed and generated.

  For additional requirements, see “Setting up the Utility Server to Send E-mail Notifications” on page 18.
- **Attach Report** - Indicates whether the report is attached to the e-mail (assuming that **E-mail Notification** is set to **Yes**).

For more information on these and other fields on the **Report Options** form, see the online help for that form.

Defining Excluded Tasks

If there are tasks which should *not* run at the same time in your system:

1. From the **Background Task Definitions** form, select a task and then click **Excluded Tasks**.
2. On the **Excluded Tasks** form, define the tasks that should not run at the same time as this task you are defining.

Creating Background Task Definitions

Note: It is necessary to create a background task only for your company's custom forms and reports. Background task definitions are already set up where needed for standard Mongoose forms, and those background task definitions are listed on the **Background Task Definitions** form. For more detail about defining background tasks, see the online help.

Use the **Background Task Definitions** form to create a record that identifies the background task to TaskMan.

Every report and every utility or activity that can be run as a background process must be listed in the **Background Task Definitions** form.

Background tasks must meet these requirements:

- The task name should match the form name to make it easier to identify.
- Executable programs must reside in a directory available to the directory where TaskMan resides.
- Reports are processed and generated using the Microsoft SQL Server Reporting Service (SSRS). The report output file must be placed in the Reports folder on the same utility server where TaskMan is installed. Although the report uses a stored procedure or IDO, its Executable Type must be set to RPT.

Note: Crystal Reports is no longer used as the default report-generating engine. Crystal Reports, up through version 2008, is still supported for backward compatibility with any custom reports you might have in your system, but SSRS is now the default engine. For information on using Crystal Reports, see your Crystal Reports documentation.

To create a background task definition:

1. Create a new record on the **Background Task Definitions** form.
2. Specify a task name (for example, DatabaseSchemaReport) and, optionally, a description.
3. Specify an executable:
   - For a report - Specify the name of the report (for example, DatabaseSchema) in the **Executable Name** field and the type of executable (RPT) in the **Executable Type** field. You must also specify in the **Report Type** field whether it is a SQL Server Reporting Services (**SSRS**) or Crystal Reports (**CR**) report.
   - For a stored procedure - Specify the procedure name in the **Executable Name** field and select **SP** in the **Executable Type** field.
   - For an executable program - Specify the program name and path in the **Executable Name** field and select **EXE** in the **Executable Type** field.
   - For an IDO - Specify the name in the **Executable Name** field and select **IDOMTH** in the **Executable Type** field.
4. Click the buttons on the form to specify any report options or excluded tasks.
5. Save the record.

Your new background task can be called from any form as an event handler.

Running Stored Procedures, Executables, and IDO Methods Using TaskMan

In addition to running reports, you can use TaskMan to run stored procedures, executables, or IDO methods. For information on using TaskMan to run reports, see *Creating and Customizing Reports*.

Stored Procedures

TaskMan can execute stored procedures directly, without going through the IDO layer. To do this, TaskMan bundles the stored procedure in a transaction.

The following example shows the steps to set up a stored procedure that runs through TaskMan from a form:

1. Use the **Background Task Definition** form to set up a record of type **SP** with the name of the stored procedure as the task name.

   For example, create a task called **AddProcessErrorLogSp**, of type **SP**.

2. Add a button to a form. Name the button **TestSP** and assign **sTest** as the caption.

   This button will be used to run the stored procedure.

3. On the **Components** property sheet for the button, create an event named **RunSp**:
   a. Click the Events button (appears as a yellow lightning bolt on the **Component** properties sheet).
   b. Click **Primary**, and then click the associated ellipses (**...**) button.
   c. In the **Event Handlers** dialog box, click **New**.
   d. In the **Event Handler Properties** dialog box, set these values:
      - **Event** - **RunSP**
      - **Description** - Enter an optional description or leave blank.
      - **Type** - **Run Background Task**
   e. For the parameters:
      - Click **Parms**, and then click the ellipses (**...**) button.
      - In the **Event Handler Parms** dialog box, for the **Error Message**, specify **mBackendMessage**.
      - For the **Success Message**, specify **sSubmitted**.
      - Click **Type Specific Parameters**.
      - In the **Edit Background Task Name and Parms** dialog box, for the **Task Name**, specify **AddProcessErrorLogSp**.
      - For the **Task Parms**, specify **BG~TASKID~,FV(TestMessage)**.

      The **FV** keyword tells Mongoose to enclose the value of **TestMessage** in single quotes. The **BG~TASKID~** substitution keyword is replaced at run time with the task number.

      For a complete list of keywords, see “Taskman Substitution Keywords” on page 26.
   f. Click **OK** repeatedly until you return to the form.
   g. Verify that **RunSP** is the primary event for the **TestSP** button.
   h. Save the form.
4. Add an **Edit** box to the form with a variable called **TestMessage** as the data source.
5. Save the form.
6. Enter a message in the **TestMessage** edit box, and submit the task.

   The system should display a message box that says: **Submitted**.

7. Open the **Background Task History** form. When the task completes, the test message should show up in the event log.

Executables

TaskMan can also be used to execute a command string as an operating system command shell. TaskMan takes the executable name from a Background Task Definitions record, appends the parameters from the Mongoose form, and attempts to execute the line.

The following example shows the steps to set up an executable that runs through TaskMan from a form:

1. Create a file called **DeletePreviewFiles.cmd** in your **TaskMan** folder. The text of this file should be as follows:

   FOR /D %%D IN ("*taskman_dir*\Report\OutputFiles*") DO del /Q "%%D\Preview*"

   where *taskman_dir* is replaced by the name of your TaskMan folder.
2. Set up a Background Task Definition record:
   - **Task Name** - **DeletePreviewFiles**
   - **Executable Name** - **DeletePreviewFiles.cmd**
   - **Executable Type** - **EXE**

   If the executable is not in the TaskMan folder, include the path to the executable in the **Executable Name** field (for example, C:\Program Files\MyExecutable.exe).
3. Add a button to a form, giving it the name **TextEXE** and the caption **sTest**.

   This button will be used to run the executable.
4. On the **Components** property sheet for the button, create an event named **RunExe**:
   a. Click the Events button (appears as a yellow lightning bolt on the **Component** properties sheet).
   b. Click **Primary**, and then click the associated ellipses (**...**) button.
   c. In the **Event Handlers** dialog box, click **New**.
   d. In the **Event Handler Properties** dialog box, set these values:
      - **Event** - **RunExe**
      - **Description** - Enter an optional description or leave blank.
      - **Type** - **Run Background Task**
   e. For the parameters:
      - Click **Parms**, and then click the ellipses (**...**) button.
      - In the **Event Handler Parms** dialog box, for the **Error Message**, specify **mBackendMessage**.
      - For the **Success Message**, specify **sSuccess**.
      - Click **Type Specific Parameters**.

- In the **Edit Background Task Name and Parms** dialog box, for the **Task Name**, specify **DeletePreviewFiles**.

f. Click **OK** repeatedly until you return to the form.

g. Verify that **RunExe** is the primary event for the **TestEXE** button.

h. Save the form.

When you click the button, this batch file will delete all the Print Preview intermediate files.

Database Connections

TaskMan will update ActiveBGTasks and BGTaskHistory for EXE tasks. It is up to the executable to handle all other database connections.

Returning Error Information

To get information back to TaskMan from an executable, use any of these methods in the EXE:

- Put a message in the ProcessErrorLog table. These messages appear in the **Task Messages** tab on the **Background Task History** form.
- Print the error message in a file called

  taskman-install-directory\Output*task-name_task-number*.txt

  TaskMan uses this as an error message in BGTaskHistory.
- Return an integer error code. TaskMan puts the EXE return code in the BGTaskHistory table, with return code 0 indicating success.

IDO Methods

This example shows the steps to set up an IDO method that runs usingTaskMan from a form:

1. Set up a **Background Task Definition** record:
   - **Task Name** - Name of IDO method
   - **Executable Name** - Enter this using the format *IDO.method*
   - **Executable Type** - **IDOMTH**
2. In a form, set up an event whose task parameters match the IDO method’s parameters. You can pass bare values (for example, "MyParameter1,MyParameter2"). Note that Product Name (variable) will not allow you to use ~LIT~ syntax as part of a value. If white spaces are significant, use the usual Product Name (variable) keywords such as P(...), V(...), C(...), and FPC(...), FV(...), or FC(...).
3. Save the form and event.

Managing Background Tasks

You might want or need to schedule some processes to run at certain times of day and at regular intervals. Use these forms to manage those processes:

- The **Background Queue** form
- The **Active Background Tasks** form
- The **Background Task Definitions** form
- The **Background Task History** form

Scheduling Reports and Utilities to Run in the Background

If a report or utility has a **Background** option on its **Actions** menu, you can schedule the report or utility to run at a time you choose. The background task can be run once, or it can be set up as a recurring task that runs at certain times daily, weekly, or monthly.

To schedule a report or utility to run in the background:

1. From the report or utility form, select **Actions > Background**.
2. In the **Background Queue** form, specify whether you want the task to run once or on a daily, weekly, or monthly basis.
3. Specify the times and dates as described in the online help.

Checking Which Background Tasks Are Running

When the **Background Queue** form is used to create a job, it creates an Active Background Task record with status of WAITING, plus the appropriate task name and parameters. It then creates a SQL Server job using the task name as the job name.

To check which background tasks are running:

1. Open the **Active Background Tasks** form.
2. Be aware that:
   - Tasks whose status is READY will process right away and cannot be deleted from the queue from within Mongoose.
   - Tasks whose status is RUNNING are currently processing and cannot be deleted from the queue from within Mongoose.
   - Tasks with a WAITING status remain in the queue until their scheduling requirements are met (as set in the **Background Queue** form). You can delete a WAITING task from the queue.

Note: Although you can’t stop a background task with a status of READY or RUNNING from within Mongoose, you can use the Windows Task Manager to do so if you need to stop a long-running process. Note that the Windows Task Manager is different from the TaskMan described in this chapter.

Deleting a Waiting task

When you delete a task on the **Active Background Tasks** form, both the record and the corresponding SQL Server job are deleted.

It is possible to use the **Background Queue** form to add a task several times. This adds job steps to the SQL job. When you delete a task with multiple job steps from the **Active Background Tasks** form, be sure to delete the correct job step.

To delete the task, mark the record for deletion and then save it.

Viewing Background Tasks That Have Run

To display information about background tasks that have been run, whether they have completed successfully or not, use the **Background Task History** form.

Return codes generated by background tasks include these:

- **Stored procedures** - Developers supply the exit status for a stored procedure within the coding of the procedure. Generally, the exit status of the stored procedure functions as its return code, which is returned by TaskMan and displayed on this form.

Note: Task Manager is a Windows service which monitors the application database in order to execute background tasks. See “Overview” on page 15 for more information on Task Manager.

- **Executable programs** - Developers supply the exit status or return code for an executable program within the coding of the program. This return code is returned by TaskMan and displayed on this form.
- **Reports** - Background tasks of the type RPT are reports. The RunReport.exe application returns exit codes to TaskMan that are displayed on this form. If the return code indicates an error, you can also consult the error log for additional information.

Using Store and Get Options with Tasks

Store Options

For any report or utility in Mongoose, you can save the information you’ve entered for later use. To do this, use the **Store Options** form:

Note: System administrators can store options for any user. Non-administrators can only store options for the current user.

1. After entering the desired information on the report or utility form, select **Actions > Store Options** to save your entries.

2. Provide the appropriate user name.
3. Create an ID in the **ID** field.

   For example, if you are storing options for the **Database Schema Report**, you might create an ID of **DBSCHR**.

4. Click **OK**.

Get Options

To recall any stored options, use the **Get Options** form:

1. After storing options for a form, from that report or utility for which you want those options, select **Actions > Get Options**.
2. Choose the appropriate user name and ID.

   The ID was set on the **Store Options** form.

3. Click **OK**.

Options Defaults

Once you have stored options for a form, you can view them on the **Options Defaults** form:

1. Open the **Options Defaults** form.
2. Enter the user name, form name, and ID.

   Note that the form name is not the same as the form title.

3. Click the Filter-In-Place button on the toolbar.

Taskman Substitution Keywords

TaskMan supports the following substitution keywords. Before executing a task, TaskMan replaces these keywords with their appropriate values when creating a string made up of the task executable name (defined on the **Background Task Definitions** form) plus the task parameters (passed from the Mongoose form).

Generally you specify these keywords when defining task parameters for an event on a form.

Some of these keywords are used to get values from TaskMan settings.

Keyword	Description
BG~TASKID~	Replaced by the task number from the ActiveBGTasks and BGTaskHistory tables.
BG~TASKNAME~	Replaced by the task name from the BGTaskDefinitions table.
BG~CONFIG~	Replaced by the Mongoose configuration name.
BG~REQUSER~	Replaced by the user name requesting the task.
BG~SQLLOGIN~	Login associated with the user ID used to connect to a database.
BG~DSN~	Name of the DSN used by TaskMan to connect to a database.
BG~UID~	User ID used by TaskMan to connect to a database.
BG~SERVER~	Server name used by TaskMan to connect to a database.
BG~DB~	Database name used by TaskMan to connect to a database.
BG~TMHOMEDIR~	Directory where TaskMan.exe and RunReport.exe are installed.
BG~ERRFILE~	When a task completes, TaskMan copies the contents of this file to BGTaskHistory, and then deletes the file.
BG~OUTDIR~	Path to the output directory under the TaskMan home directory. This allows EXEs run through TaskMan to produce output.
BG~ISOLATIONLEVEL~	The SQL Server isolation level used by TaskMan for reports and stored procedures. The return value is UNCOMMITTED or COMMITTED. Values are set in the Isolation Level field on the **Background Task Definitions** form and the **Collection Read Mode** field on the **Process Defaults** form.
BG~FAXNUM~	The telephone number of the fax machine to which the document is to be sent.
BG~FAXSERVER~	The name of the fax server machine. The keyword applies only to Windows Fax and Infor Framework Fax Service. If BG~FAXSERVER~ is omitted from a fax header or if the Fax Server field on the **Intranets** form is blank, then the default fax server name at run time is: ■ If Windows Fax is configured, then the server name is the name of the TaskMan machine. Windows Fax uses a modem on the TaskMan machine. ■ If Infor Framework Fax Service is used, the server name is the name of the machine on which the service runs.
BG~FAXTOCOMPANY~	The name of the company to which the fax is sent; the name is printed on the fax cover sheet. The keyword does not apply to Windows Fax or Infor Framework Fax Service.
BG~FAXTONAME~	The name of the individual to whom the fax is sent; the name is printed on the fax cover sheet. The keyword does not apply to Windows Fax or Infor Framework Fax Service.
BG~OUTPUTFILE~	The report output file created in the directory \<*TaskMan_Directory*>\Report\OutputFiles\<*user*> on the TaskMan machine. With Infor Framework Fax Service, the file is accessed directly from this directory.

TaskMan also supports the following additional keywords used as command line switches. TaskMan deletes these keywords from the task executable and parameter string.

Keyword	Description
BG~LEAVELITS~	Mongoose wraps many literal values in ~LIT~(...). This keyword, which can be specified in an event handler on a form, tells TaskMan to leave these values. For example, you can create an event on a form that runs an EXE through TaskMan. Use this keyword to allow any ~LIT~ keywords in the parameters to be passed through to the EXE.
BG~LEAVETEMPS~	TaskMan normally puts the contents of the BG~ERRFILE~ error file in BGTaskHistory and then deletes the file. This keyword tells TaskMan to leave the error files.

Running TaskMan in Debug Mode

If you are having problems with a background task, you can run TaskMan in debug mode. Doing so generates additional messages for the Microsoft Event Viewer.

Enabling Debug Mode On the **Process Defaults** Form

By using the **Process Defaults** form, you do not have to stop and restart the Infor Framework TaskMan service. For more information about the **Process Defaults** form, see the Mongoose online help.

1. Open the **Process Defaults** form.
2. In the **Process Name** field, select **TaskMan Options**.
3. In the **Default Value** field, enter **debug**.
4. Save the record and exit the form.

Enabling Debug Mode By Restarting the Service

To enable debug mode by restarting the Infor Framework TaskMan service:

1. On the server where TaskMan resides, open Windows Services.
2. In the list of services, select Infor Framework TaskMan.
3. If the Infor Framework TaskMan service is running, stop it.

Caution: When you stop TaskMan, all running tasks are terminated.

4. Right-click on the Infor Framework TaskMan service and select **Properties**.
5. In the **Properties** dialog box **Start parameters** field, enter **debug**.

6. To restart TaskMan, click the **Start** button.

Caution: When you finish debugging the problem, be sure to stop the TaskMan service, remove the debug parameter, and restart it.

TaskMan debug mode messages are listed and explained starting on page 29.

There is also an optional "nowait" parameter for TaskMan. If you are starting TaskMan manually, this keyword allows it to start faster.

RunReport debug mode messages are listed and explained in the *Reporting Guide*.

TaskMan Debug Mode Messages

<***SPname***> After Call <***Taskman source file***>: <***Source file line number***>

This message is printed after a stored procedure is called.

Active Task Set not open: <***DatabaseInfo***>. <***Taskman source file***>: <***Source file line number***>

TaskMan is trying to clear database connections.

Cannot find last slash. <***Taskman source file***>: <***Source file line number***>

Error while retrieving information about the home directory from which TaskMan is executing.

Close process connection completed for Task <***n***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Closing database: <***dsn***>.<***Taskman source file***>:<***Source file line number***>

Informational message.

Closing process connection for Task <***n***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Decrement <***taskname***> <***tasknumber***>. <***Taskman source file***>:<***Source file line number***>

Informational message - TaskMan is decrementing the Running Tasks list when the task completes.

DELETE ActiveBGTasks where TaskNumber = <***n***>. <***Taskman source file***>:<***Source file line number***>

Informational message - the task is deleted from the active tasks table.

Deleting TaskInfo handle. <***Taskman source file***>:<***Source file line number***>

Informational message - TaskMan is cleaning up report (RPT) process handles.

Error retrieving TaskMan Module Name. <***Taskman source file***>:<***Source file line number***>

TaskMan retrieves its module name in order to get its home directory. There was an error in retrieving the module name, so TaskMan cannot determine its home directory.

Increment <***taskname***> <***tasknumber***> Total requests <***n***> Queue size <***size***>. <***Taskman source file***>:<***Source file line number***>

TaskMan is polling the table for active background tasks in the each configured application database.

No Intranet records found, using defaults. <***Taskman source file***>:<***Source file line number***>

TaskMan is using the default values because it could not find a matching Intranets record.

Opening Intranet record set. <***Taskman source file***>:<***Source file line number***>

Informational message.

Pause to ensure SQL Server is completely up. <***Taskman source file***>:<***Source file line number***>

TaskMan pauses to make sure that SQL Server has started before trying to access databases.

Poll=<***n***> Connect=<***n***> Process=<***n***> MaxNo=<***n***> NumRec=<***n***> site = <***site***> Intranet = <***intranetname***> String Table = <***stringtable***> URL = <***URLpath***> Format = <***outputformat***> ReportPath = <***path***> Email Notif = <***emailnotification***>. <***Taskman source file***>:<***Source file line number***>

TaskMan successfully queried the Intranet table and retrieved the information listed here.

Rpt task failed. <***Taskman source file***>:<***Source file line number***>

TaskMan failed while trying to run a report task.

ServiceMain starting. <***Taskman source file***>:<***Source file line number***>

Informational message.

SQLCancel failed. Deleting TaskInfo handle. <***Taskman source file***>:<***Source file line number***>

The SQL Cancel of the process and task was not successful.

SQLCancel of Task <***tasknumber***> completed. Return Code = <***n***>. <***Taskman source file***>:<***Source file line number***>

TaskMan canceled a stored procedure background task.

Task <***tasknumber***> <***taskname***>. An error occurred while waiting for the process to finish. Error return = <***code***> message = <***message***>. <***Taskman source file***>:<***Source file line number***>

A Windows error occurred.

Task <***tasknumber***> <***taskname***>. Call to AddProcessErrorLogSp failed for user <***userID***>. Return code = <***code***>. <***Taskman source file***>:<***Source file line number***>

TaskMan tried and failed to add a task message to **Background Task History**.

Task <***tasknumber***> <***taskname***>. Call to CloseSessionSp failed for task <***taskname***> user <***userID***>. Return code = <***code***>, Error message = <***message***>. <***Taskman source file***>:<***Source file line number***>

TaskMan could not retrieve the Report Options information.

Task <***tasknumber***> <***taskname***>. Call to GetTaskOptionsSp failed for task <***taskname***> user <***userID***>. Return code = <***code***>, Error message = <***message***>. <***Taskman source file***>:<***Source file line number***>

Task <***tasknumber***> <***taskname***>. Call to InitSessionContextSp failed for task <***taskname***> user <***userId***>. Return code = <***code***>, Error message = <***message***>. <***Taskman source file***>:<***Source file line number***>

Task <***tasknumber***> <***taskname***>. Calling AddProcessErrorLogSp. <***commandline***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Calling sp.Call <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Can transact. <***Taskman source file***>:<***Source file line number***>

TaskMan could not start a stored procedure background task.

Task <***tasknumber***> <***taskname***>. Cannot transact. <***Taskman source file***>:<***Source file line number***>

TaskMan cannot start a transaction for a stored procedure task. The stored procedure is not executed.

Task <***tasknumber***> <***taskname***>. CloseSessionSp called with Input Parameter <***sessionID***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Could not close session <***sessionID***>. <***Taskman source file***>:<***Source file line number***>

Task <***tasknumber***> <***taskname***>. Creating directory <***directory***>. Result = <***code***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Database not open. Cannot enter Process Error: <***message***>. <***Taskman source file***>:<***Source file line number***>

Task <***tasknumber***> <***taskname***>. DELETE ActiveBGTasks where TaskNumber = <***n***>. <***Taskman source file***>:<***Source file line number***>

Informational message - the task is deleted from the active tasks table.

Task <***tasknumber***> <***taskname***>. Error <***code***> <***message***> when starting Command - <***commandline***>. <***Taskman source file***>:<***Source file line number***>

An error occurred when TaskMan was starting the specified command.

Task <***tasknumber***> <***taskname***>. Error moving file <***outputfile***> to <***new outputfile***>. <***Taskman source file***>:<***Source file line number***>

TaskMan could not copy the output file to the **OutputFile** folder. Be sure permissions are set up properly, and the folder exists.

Task <***tasknumber***> <***taskname***>. Exiting RunTask: Removing Task from List. <***Taskman source file***>:<***Source file line number***>

The background task has finished running.

Task <***tasknumber***> <***taskname***>. Fax=<***faxname***>, Fax Server = <***faxserver***>, Output Format=<***outputformat***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. GetExitCodeProcess return = <***n***>. <***Taskman source file***>:<***Source file line number***>

This message displays the return code for an executable program background task.

Task <***tasknumber***> <***taskname***>. GetTaskOptionsSp called with Input Parameters <***taskname***>, <***userID***>, <***stringID***> returned Output Parameters format = <***output format***>, printer = <***printer name***>, email = <***email notification***>, attach = <***attach report***>, email address = <***email address***>, Return code = <***code***>, Error message = <***message***>, String Table = <***string table***>, Fax Server = <***fax server***>.<***Taskman source file***>:<***Source file line number***>

TaskMan is retrieving specific information about the run-time user requesting this report background task. This information is entered in the **Report Options** and **Intranets** forms.

Task <***tasknumber***> <***taskname***>. InitSessionContextSp called with Input Parameter <***taskname***>, returned Output Parameter <***sessionID***>, Return code = <***code***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. InitSessionContextSp failed. Could not call SP. <***Taskman source file***>:<***Source file line number***>

Task <***tasknumber***> <***taskname***>. No attachment sent: either the report was sent to the printer, the Task Type was not RPT, or the report didn't complete successfully. <***Taskman source file***>:<***Source file line number***>

Task <***tasknumber***> <***taskname***>. Opening user names failed. <***Taskman source file***>:<***Source file line number***>

TaskMan encountered a problem attempting to open the UserNames database table.

Task <***tasknumber***> <***taskname***>. Password decrypted. <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Profile Values -- String Table: <***stringtable***>, String ID: <***stringID***>, Fax: <***fax name***>, Email: <***email address***>, Number of copies: <***n***>, Printer: <***printer name***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Return = <***code***> WaitForSingleObject return = <***code***>. <***Taskman source file***>:<***Source file line number***>

This message is entered immediately after a background task has ended.

Task <***tasknumber***> <***taskname***>. Running: <***stored procedure***>.<***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Running EXE: <***command line***>.<***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Running Report: <***RunReport.exe command line***>.<***Taskman source file***>:<***Source file line number***>

This is the command line TaskMan executes to run a report. For debugging purposes, you can paste the following command line into a batch file and execute it:

```
RunReport.exe <Command line>
```

Task <***tasknumber***> <***taskname***>. Sending Fax: <***command line***>

Informational message.

Task <***tasknumber***> <***taskname***>. Setting Connection Timeout <***n***> and opening a connection for this task. <***Taskman source file***>:<***Source file line number***>

TaskMan opened a connection for this background task that will be used to update the Task History table and to delete entries from the Active Task table in the application database.

Task <***tasknumber***> <***taskname***>. Setting Process Timeout <***n***>.<***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. Stored Procedure call GetSQLServerLoginSp failed with return code <***code***> and error message <***message***>. <***Taskman source file***>:<***Source file line number***>

TaskMan could not get the SQL login information needed to process the task.

Task <***tasknumber***> <***taskname***>. Successful termination of Task. Performing commit. <***Taskman source file***>:<***Source file line number***>

A stored procedure background task can be committed.

Task <***tasknumber***> <***taskname***>. Unable to access <***document path***>. Fax not sent. <***Taskman source file***>:<***Source file line number***>

TaskMan could not access the RTF file using the specified path. Check to see that the file and path exist and that permissions allow TaskMan access to it.

Task <***tasknumber***> <***taskname***>. Unable to access <***SendFax.exe path***>. Fax not sent. <***Taskman source file***>:<***Source file line number***>

TaskMan could not access the Fax.exe utility using the specified path. Check to see that the file and path exist and that permissions allow TaskMan access to it.

Task <***tasknumber***> <***taskname***>. UPDATE ActiveBGTasks SET TaskStatusCode = 'RUNNING' where TaskNumber = <***n***> .<***Taskman source file***>:<***Source file line number***>

As TaskMan selects a background task to run, it changes its status to running.

Task <***tasknumber***> <***taskname***>. UPDATE BGTaskHistory SET CompletionDate = <***date***>, CompletionStatus = <***status***>, TaskErrorMsg = '<***message***>' where TaskNumber = <***n***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Task <***tasknumber***> <***taskname***>. UPDATE BGTaskHistory SET ProcessId = <***pid***> WHERE TaskNumber = <***n***>. <***Taskman source file***>:<***Source file line number***>

This message displays the process ID (pid) for a background task. You can use the process ID to trace the status of a background task in the Windows TaskMan. This ID is also displayed in the **Background Task History** form.

Task <***tasknumber***> <***taskname***>. UserNames.Username = '<***userID***>'.<***Taskman source file***>:<***Source file line number***>

The user ID of the run-time user requesting the current background task.

Task <***tasknumber***> <***taskname***>. Usernames.Username = <***userID***> Groupname = <***groupname***>.<***Taskman source file***>:<***Source file line number***>

The user ID and group name of the run-time user requesting the current background task.

Taskman Error: TaskCounter is not keeping accurate count of Running Tasks. <***Taskman source file***>:<***Source file line number***>

The number of running tasks in the system does not match the task counter value.

Taskman home directory: <***homedirectory***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

TaskMan Stopping: Clearing database connections. <***Taskman source file***>:<***Source file line number***>

Informational message.

The maximum number of concurrently running tasks is <***n***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

<***dsn***> UID <***userID***> PWD **** String Table=<***stringtable***> Poll=<***n***> Process Timeout=<***n***> Connection Timeout=<***n***> Max Num Tasks=<***n***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

UPDATE BGTaskHistory SET CompletionDate = <***date***>, CompletionStatus = <***status***>, TaskErrorMsg = '<***message***>' where TaskNumber = <***n***>. <***Taskman source file***>:<***Source file line number***>

Informational message.

Checking Infor Framework TaskMan Events in the Event Log

In addition to the debug messages mentioned above, TaskMan generates event messages during normal processing. You can view these messages in the Microsoft Event Viewer. Some of the more common messages are listed and described in “Event Messages from TaskMan” on page 42.

To access the Event Viewer, follow these steps on the server where TaskMan resides:

1. Select **Settings > Control Panel** from the Windows Start menu.
2. Open **Administrative Tools** and then **Event Viewer**.
3. Select **Application Log**.

Troubleshooting

This section describes possible problems with TaskMan, and how to solve them. Additional problems and solutions that are specific to reports are provided in the *Creating and Customizing Reports* guide.

TaskMan Does Not Start

Symptoms

The Infor Framework TaskMan service is not starting, and you see this message in the Application Event log:

> No database definitions defined. TaskMan must be configured before starting the service.

Possible Solutions

- Run the Server Service Configuration utility and verify that the correct application databases are configured for TaskMan.

- Infor Framework TaskMan for Mongoose might be running under a user ID that does not have privileges to access the registry. Try restarting the service to run as a local system account. If it starts, then the problem is with the user ID.
- In Windows, restart Infor Framework TaskMan as an active service.

Changes to **Intranets** Form Settings Are Ignored

Symptoms

Changes are made on the Intranets form, but those changes are not reflected in tasks that subsequently run that should be using those changed values.

Possible Solutions

When the Infor Framework TaskMan service starts, it caches information from the Intranets table. If you then change a setting on the **Intranets** form, TaskMan does not see this change. Instead, it continues to use the cached setting for the following fields:

- Polling Interval
- Connection Query Timeout
- Process Timeout
- Maximum Concurrent Tasks

For this reason, after changing one of these settings on the **Intranets** form, you must stop and restart the Infor Framework TaskMan service so the change can take effect.

Background Task Runs But Has No History Record

A background task is submitted from a Mongoose form. It appears to have been submitted without any problems, but no record is created in the **Background Task History** form. The BGTaskHistory record is created by a trigger on the ActiveBGTasks table. ActiveBGTasks is the queue of tasks submitted to TaskMan. So, if there is no history record, the task never made it to the queue, despite any messages that might have displayed on the Mongoose form.

This is probably a bug in the sequence of form events the Mongoose form used to submit the task.

Labels Not Replaced with String Table Values

Symptoms

Component labels are not being replaced with string table values or are not being translated properly.

Possible Solutions

Use the Site Name from the **System Parameters** form to select the correct record on the Sites form. Make sure that, for this site, there is a value in the **Sites** form's **Forms Database Name** field (in some versions, labeled as the **Strings Table Specification** field). If the forms database is on a different server than the application database, the field's value should also indicate the linked server name, in this format:

server_name.Forms_database

Infor Framework TaskMan then determines the proper strings table name by searching the specified forms database for the strings table associated with the current Product Name (variable) session.

If labels are not translated, stop and restart TaskMan. The **Strings Table Specification** field (or **Forms Database Name** field) might have been modified after TaskMan was last started. TaskMan checks this value only once, when it first starts up.

The Transport Failed to Connect to the Server

Symptoms

When you attempt to send an e-mail with the SMTP protocol, or if you receive a similar error when testing the SMTP protocol using Telnet, TaskMan returns the error:

The transport failed to connect to the server.

Possible Solutions

The firewall or antivirus software on the mail server might be blocking the e-mail.

For more information on setting up e-mail notifications, see "Setting up the Utility Server to Send E-mail Notifications" on page 18.

Event Messages from TaskMan

TaskMan runs as a service under Windows, called Infor Framework TaskMan, and generates event messages that you can view in the Microsoft Event Viewer. If you are having problems with a background task, you can run TaskMan in debug mode (see page 28), which generates additional messages for the Microsoft Event Viewer.

The following messages are generated normally and do not require Infor Framework TaskMan to be running in debug mode. If a database exception occurs, TaskMan tries to retrieve and log the error message.

<***function name***> failed with return code <***code***> in <***Taskman source file***> at <***Source file line number***>.

Abnormal termination of Task <***n***> (returncode = <***code***>). Performing rollback.

A stored procedure background task was rolled back due to one of the following circumstances:

- The stored procedure generated a return code less than 0 or greater than 5.
- Executing the stored procedure generated an exception.

An unknown exception occurred. <***Taskman source file***>: <***Source file line number***>

Canceling Process <***taskname***>, Task <***n***>

If TaskMan is shut down, an event log message will be printed as each running background task that is a stored procedure is canceled.

Canceling Report <***taskname***>, Task <***n***>

If TaskMan is shut down, an event log message will be printed as each running background task that is a report is canceled. Some reports might be left hanging even if TaskMan is shut down.

DSN <***dsn***> Database <***db***>. Login successful

Informational message.

Either the DSN or the SQL login was not set. <***ReturnMessage***>

Informational message.

RegisterServiceCtrlHandler failed with return code <***n***> in <***Taskman source file***> at <***Source file line number***>.

RSTaskMan shutdown: <***n***> tasks are running and will be canceled.

If TaskMan is shut down while tasks are running, these tasks will be canceled.

RSTaskMan starting: Version <***n***>

Informational message.

RSTaskMan Terminating

Informational message.

RSTaskMan Terminating - TaskMan Home Directory not properly detected

TaskMan could not determine its home directory and will shut down.

SetServiceStatus failed in <***Taskman source file***> at <***Source file line number***>.

StartServiceCtrlDispatcher failed with return code <***n***> in <***Taskman source file***> at <***Source file line number***>.

TaskMan was not able to start the Service Dispatcher.

TaskTypeCode not recognized

The task type was not EXE, RPT, SP, or IDOMTH.

Using SQL Profiler to Trace TaskMan Instances

To use SQL Profiler to see which instance of TaskMan is monitoring an application database:

1. Start the SQL Server Profiler.
2. In SQL Profiler, select **File > New Trace**.
3. Connect to the application database server you want to trace.
4. On the Events Selection tab, remove everything from the **Events** column except **TSQL -- SQL:Batch completed**.
5. On the **Data Columns** tab, add **Server Name** to the **Selected Data Column**.
6. To start the trace, click **Run**.

Stored Procedures Used for Performance Benefit

These are stored procedures you should run regularly for performance benefits:

- **sp_updatestats** - Run this SQL-supplied stored procedure regularly for statistics updating for all tables in a database. You can get more information about this stored procedure from the SQL online help.
- **SLServerRestartSp** - This stored procedure runs whenever the database server is restarted (since no one is logged in at that time) and performs general cleanup.
- **PurgeNextKeysSp** - Run this stored procedure to clean up the NextKeys table. NextKey records are inserted, never updated to get concurrency. This stored procedure cleans out the extra rows. DO NOT run this utility while others are using the system. The utility will lock users out, but you should log everyone out of the system before running this utility.

Chapter 4: License Management

4

As System Administrator, you need to manage your license agreement, and Mongoose allows you to do that. This chapter tells you how to apply your license and how to perform a post license application step if necessary.

There is much more to know about managing your license. For additional information on license management, see the guide entitled *Infor Mongoose Licensing Guide* on www.infor.com/inforxtreme.

License Management Form

This form allows you to manage your company's license agreement. The form shows when the license begins and ends, which modules are licensed, and it is where you apply the license to stay current.

Apply a License

1. On the **License Management** form, click the **Apply License** tab.
2. Paste your license document into the **License Document** field. You can cut and paste the information from the license document into this field, or you can select the **Browse** button and select the document from your drive. The document is encrypted and every character counts, so if you copy and paste, make sure to copy the entire document.
3. Once you have entered the license document, click the **Apply License** button. Any previous multi-session users that you had are deleted and replaced with the ones defined in the new license document.

Note: Let us assume you are applying a new license to replace an old one for the MGCoreTrans module. The old license was for 50 users, and the new one is for 40. When you click the Apply License button, you get an error message. If the new license is for fewer users than the old license, you need to go to the **Users Module** form and delete users' association with the MGCoreTrans module until you have the same amount or fewer than the new license allows. If your new license is for the same number of users or more, then you can apply the license and you don't need to do anything else.

Post License Application Steps

Note: You must follow the steps in this section if there were active WinStudio sessions when you applied the license. If there were not any active sessions, you can skip this step. You can perform these steps, however, in either case.

1. On the utility server, open the **Configuration Manager**.
2. Select the **Utilities** tab.
3. Use the **Utilities** tab to discard the Runtime Service Cache for all configurations that reference the application database to which the new license has just been applied. See the Configuration Manager online help for more information about the options on this tab.

Chapter 5: Multi-Site

5

In Mongoose, each site's database does not have to be connected to any other database in order to function. Data is transferred through a process called replication. The system manages its own replication which is different from SQL's replication. We do not support standard SQL replication as a means to transfer data from one site to another.

The connection between sites can be either:

- **Transactional** - This is a constant live connection from one site to another. Any update is made to the target site immediately. The down side is that if the target site is down for any reason, the transfer of data does not occur;

  Or

- **Non-transactional** - There is no live connection between sites. If using a non-transactional (also referred to as delayed replication or asynchronous replication) setup, data that is shared must be replicated.

For more detailed information on replication and multi-site use, see these documents on www.infor.com/inforxtreme:

- *Replication Reference*
- *Multi-Site Planning Guide*
- *Multi-Site Implementation Guide*

Chapter 6: Authorizations

6

You may limit user access to certain forms or limit what they can do on certain forms by way of authorizations. Use the following four forms to manage these authorizations:

- Object Authorizations for User
- Object Authorization for Group
- Users
- Groups

Object Authorizations For User

Use this form to set up form-level security for a user or to set up security for a user based on a middle-tier IDO. An object in this case is either the name of the form or the name of the IDO.

If the user specified on this form is part of a group, the user authorizations that you set up on this form override any group authorizations that apply to the same user.

You can open the form right from the Explorer window, or you can click the **User Authorizations** button on the **Users** form.

The following privileges can be granted or revoked for the user for a specified form in the Object Authorizations For User form:

- **Delete** controls whether the user can delete records.
- **Edit** controls whether the user can edit existing data.
- **Execute** controls all privileges, including whether the user can open the form. When execute is revoked, all privileges are unavailable.
- **Insert** controls whether the user can insert records.
- **Read** controls whether the user can read data.
- **Bulk Update** controls whether the user can perform a bulk update, such as multiple find and replace operations.
- **Update** controls whether the user can save changes to existing data.

Object Authorization for Group

Use this form to set up form-level security for a group of users or to set up security for a group based on a middle-tier IDO.

Any user authorizations for individuals will override the group authorizations defined on this form.

You can open the form right from the Explorer window, or you can click the **Group Authorizations** button on the **Groups** form.

The following privileges can be granted or not granted for the group in the Object Authorization For Groups form:

- **Delete** controls whether the user can delete records.
- **Edit** controls whether the user can edit existing data.
- **Execute** controls all privileges, including whether the user can open the form. When execute is not granted, all privileges are unavailable.
- **Insert** controls whether the user can insert records.
- **Read** controls whether the user can read data.
- **Update** controls whether the user can save changes to existing data.
- **Bulk Update** controls whether the user can perform a bulk update, such as multiple find and replace operations.

Users

Use this form to register users to the application. A user ID is required for each user who logs on to the application. You can also specify the following:

- Passwords needed to log on to the application
- Workstation IDs so users bypass the logon dialog box
- E-mail addresses so notifications can be sent about automated tasks
- Editing permissions that determine whether users can enter edit mode to create or customize forms
- Security authorizations for this user at a form level or a component level

- Groups to which the user belongs and security authorizations for that group
- Additional information about the user that the application needs.

Note: Initially, only the supplied default system administrator user ID can create or delete other user IDs.

Any user who is added to the System Administration group or designated as a super user can access the Users form and change the password for any other user. Users can change their own passwords on the **User Information** form.

Groups

Use the Groups form to create groups and to assign user IDs to them. These groups can be organized in any way that makes sense for your company or organization -- by location, by organization, by job description, and so on.

Caution: Although the system allows you to modify or delete the default groups, doing so may cause future conversion problems while upgrading and other problems. We recommend you copy the records from the default group to a new group name and modify that. Do NOT delete or modify default groups.

Assign users to groups so you can then create group authorizations that apply to every individual/ user ID in the group.

Create a Super User

A Super User can run all forms and perform all actions on all forms for which they hold a license. In some cases, actions may have to be performed by the ’sa’ user account.

If you create a Super User, you do not need to set any other authorizations for this user. The Super User status overrides all other types of authorizations.

1. Open the **Users** form.
2. Select the desired User ID.
3. Select the **Super User** field.
4. Save the record.

Assign a User to a Group

1. Open the **Users** form.

2. Select the desired User ID.
3. On the **Groups** tab, select the **Group Name** from the drop down list.
4. Save the record.

Note: When you assign a user to a group (for example, user Bob is assigned to the End User Group), that user gets access to all the forms associated with that group. If you want user Bob to have most of the access associated with that group but not all, you need to edit the user authorizations for user Bob. See “Edit Authorizations for a User in a Group” on page 53.

Assign a User to a Primary Group

In general, it is easier to first create the group, then assign user authorizations. This allows you to assign each user a Primary Group that specifies authorizations without going through each user's authorizations form by form.

1. On the **Users** form, select the desired User ID.
2. On the **Groups** tab, select the group name from the Group Name drop down list.
3. Select the **Primary Group** field.

   When selected, this field indicates that the corresponding user group is a primary group. The system uses the group designated as the Primary Group to load the correct version of customized forms for the selected user.

   Users may belong to more than one group, but only one group may be designated as the Primary Group, and only the Primary Group is used for loading group versions of a customized form. For example, suppose user Bob is a member of the End User and Developer groups with the End User group marked as Bob’s primary group. If Bob launches a form, the system looks for a group-level customized version of that form for the End User group, not the Developer group. If such a customized version exists, and if there isn't also a user-level customized version for user Bob, then that is the form the system will display.

View Group Authorizations

1. Open the **Groups** form.
2. Select the desired group name.
3. Click the **Group Authorizations** button. The **Object Authorization for Group** form opens.
4. In the **Object Name** field, select an object. The privileges are listed in the columns to the right, and those privileges are either granted or not granted.

Edit Authorizations for a User in a Group

User authorizations override group authorizations. So for example, you might want user Bob to have all of the authorizations of the End User group, except for access to the Audit Log form.

1. Assign a user to a group on the **Users** form (See “Assign a User to a Group” on page 51).
2. Open the **Users** form.
3. Select the user whose authorizations you want to edit.
4. Click the **User Authorizations** button. The **Object Authorizations for User** form opens.
5. In the **Object Name** field, select the object you wish to edit for this user. In this example, it is AccountingPeriods.
6. Depending on what you want, change the desired privileges either to **Granted** or **Revoked**.

Forms Security

Form level security is checked when a user opens a form. The AccountAuthorizations table is queried to see what privileges have been granted to the user or to the group to which the user belongs.

Change Passwords

Any user who is added to the System Administration group or designated as a super user can access the **Users** form and change the password there for any other user. Users without such authorization can change their own passwords on the **User Information** form.

Copy User Tables

If you upgrade from one version of Mongoose to another version of Mongoose, you can copy your user and group authorizations to the new version with the Copy User Tables utility so you don't have to set them up again.

Requirements

- This utility should only be used when first setting up a database for use and NOT after users have already been defined and Foreign Key references have been added.
- The latest version of Mongoose must be installed and licensed before running this utility.

Note: If your latest license is for fewer users than your old license, not all users will carry forward. When you complete the procedure, open the Users form in the new Mongoose version to check the users. You can then make changes accordingly.

1. On the Utility Server, for Windows Server 2008, select **Start > Programs > Infor > Tools > Copy User Table**. For Windows Server 2012, the Copy User Table is found under Apps.
2. On the **Source** tab, enter the following information:
   - **SQL Server Name** - Enter the name of the SQL Server or select it from the list.
   - **SQL Server Login ID** - Enter your SQL Server Login ID.
   - **SQL Server Logon Password** - Enter your SQL Server Logon Password.
   - **Use Windows Authentication** - Select this field to use Windows Authentication. This means that instead of supplying a SQL Server login/password, the server uses the username of the person currently logged in to the Windows machine and logs in automatically.
   - **Source Database** - Select the name of the application database with the user/group authorizations you want to copy.
3. Click the **Target** tab.
4. On the **Target** tab, enter the following information:
   - **SQL Server Name** - Enter the name of the SQL Server or select it from the list.
   - **SQL Server Login ID** - Enter your SQL Server Login ID.
   - **SQL Server Logon Password** - Enter your SQL Server Logon Password.
   - **Use Windows Authentication** - Select this field to use Windows Authentication. This means that instead of supplying a SQL Server login/password, the server uses the username of the person currently logged in to the Windows machine and logs in automatically.
   - **Target Database** - Select the name of the application database to which you want to copy the user/group authorizations.
5. Click the **SyteLine Login** tab.
6. Enter the following information:
   - **User ID** - Enter your SyteLine user ID.
   - **Password** - Enter the SyteLine password associated with the user ID.
   - **Data Source Config** - Select the target application database from the drop down list.
7. Click the **Copy** button. The status bar shows the progress. The process can take several minutes.

Note: In the Mongoose directory on the utility server, is a file named UserTables.txt. This file lists the tables that are copied by the Copy User Table utility.

Chapter 7: Setting up an Audit Log

7

Use the audit log to see which user performed what action at what time and on what form.

Use the following three forms to manage your audit logging:

- Process Defaults
- Audit Log Types
- Audit Log.

Process Defaults

Use the **Process Defaults** form to enable audit logging.

Set the Default Value field to **1** for Enable Audit Logging. A default value of **0** turns the audit log functionality off.

Audit Log Types

Use the **Audit Log Types** form to create types of messages to include in the audit log. Developers can create other types of messages that are generated by event handlers of the Add Entry to Audit Log response type.

The **Audit Log Types** form displays the following information:

- **Message Type** - The Message Type field indicates the type of message in the audit log. By default, messages about users logging on are 1, messages about users opening forms are 2, and messages about users modifying forms or objects are 3. If the Application Event System performs audit logging from action type Audit that is type 4, and if a user login failure occurs, that is type 5. Custom messages are numbered 10,000 and greater.
- **Category** - Category identifies the type of auditing. Three values can appear in this field:
  - **System** - Types of events are User Login and Open Form. You cannot select a category of System.

  - **Custom Form Event** - Custom Form Event allows programmers to add audit logging to any event handler; for example, when a user selects a tab on a form.
  - **Database** - Database displays when records are added, updated, and deleted, and the message displays 1) Database: Add, 2) Database: Update, or 3) Database: Delete, depending on the action performed.
- **Table or Description** - For System and Custom Form Events, this field describes the event being logged. You can change the contents of this field at any time. The value in this field appears in the Audit Log.

  For Category Database, this field indicates the table that will have an audit log.
- **Fields** - Fields is used only if the category is Database. This can be a specific database field (cust_num, for example) or an asterisk (*). Entry of an asterisk indicates that all fields in the table should be entered in the audit log.

Audit Log

Use the **Audit Log** form to view and delete messages in the audit log.

The Audit Log form displays the following information:

- **Log Description** - This field describes the type of audit log.
- **Message Type** - The Message Type field indicates the type of message in the audit log. By default, messages about users logging on are 1, messages about users opening forms are 2, and messages about users modifying forms or objects are 3. If the Application Event System performs audit logging from action type Audit that is type 4, and if a user login failure occurs, that is type 5. Custom messages are numbered 10,000 and greater.
- **Category** - Category identifies the type of auditing. Three values can appear in this field:
  - **System** - Types of events are User Login and Open Form. You cannot select a category of System.
  - **Custom Form Event** - Custom Form Event allows programmers to add audit logging to any event handler; for example, when a user selects a tab on a form.
  - **Database** - Database displays when records are added, updated, and deleted, and the message displays 1) Database: Add, 2) Database: Update, or 3) Database: Delete, depending on the action performed.
- **Message Description** - The message description is a description of the message type. The description will be one of three vales:
  - **User Login**
  - **Open Form**
  - **Custom Message**
- **Primary Key** - Primary Key identifies the database record being changed. It contains the fields that make up the primary key. When multiple fields make up a key, the values are concatenated and separated by dashes.
- **Old Value** - Old Value is only used for Category Database. Based on the activity - Add, Update, or Delete - the following is displayed:

- **Add**: *Blank*
- **Update**: *Value prior to update*
- **Delete**: *Value prior to delete.*

- **New Value** - New Value is only used for Category Database. Based on the activity - Add, Update, or Delete - the following is displayed:
  - **Add**: *Value after the add*
  - **Update**: *Value after the update*
  - **Delete**: *Blank.*
- **User Name** - The user ID whose actions generated the audit log message is displayed.
- **Date/Time** - Date/Time displays the system date and time that the audit log entry was made.

Chapter 8: Database Name Change

8

If for any reason you have changed the name of your application database, you must also change a pointer in the system to that new database name.

- **Sites form** - On the System Info tab, change the Database Name field to the new name.

Appendix A: Web Rendering Differences

A

The information formerly contained within this appendix has been moved to the online help. Most of the information is included in the topic “About Web Clients.” Other information can be found by performing a full text search of the help.

Appendix B: Performing Configuration Tasks in a Command Line

B

You can use the command line utilities in this chapter to perform configuration tasks in Mongoose.

Command Line Utilities

InforDBCL.exe

This utility contains all configuration tasks involving databases. The available commands with some examples are:

InforDbCL.exe CreateDb

-databasename: SQL database name

-username: SQL server user name

-password: SQL server password

-servername: SQL server name

-dbtype: Database type {App, Objects, Forms, Templates }

-product:Mongoose product (e.g. "Mongoose", "SL") which corresponds to the registry key

[-collation]: SQL collation to be used for the database

[-clouddb]: Indicates whether DB is already created and just needs populated

[-dbsubtype]: App database subtype (e.g. 'Demo' or 'Empty')

[-log]: full path to a log file to redirect to

[-siteid]: App database site id

[-sitename]: App database site name

[-sitedesc]: App database site description

[-sitegroup]: App database site group

[-sitecurrency]: App database site currency

[-sitetype]: App database site type {S,E}

[-sqlservertimezone]: SQL Server timezone (standard name)

[-timezone]: App database timezone (standard name)

- createdb - Use this command to create a new database.
  Examples:
  ```
  infordbcl.exe createdb -databasename:CmdApp -username:sa -password:test123
  -servername:win2012 -dbtype:App -product:SL

  infordbcl.exe createdb -databasename:CloudApp -username:sa -
  password:test123 -servername:win2012 -dbtype:App -product:SL -clouddb

  infordbcl.exe createdb -databasename:Mongoose_Forms -username:sa -
  password:test123 -servername:win2012 -dbtype:Forms -product:Mongoose
  ```

InforDbCL.exe PatchDb

-databasename: SQL database name

-username: SQL server user name

-password: SQL server password

-servername: SQL server name

-dbtype: Database type {App, Forms, Templates } The 'Objects' type is not available when patching.

-product: Mongoose product (e.g. "Mongoose", "SL") which corresponds to the install subfolder

-config: Mongoose configuration containing the App database and it's referenced Objects database to be patched. Only required when the Objects database needs patched.

[-siteid]: App database site id. Used only when the Objects database doesn't need patched.

[-log]: full path to a log file to redirect to

- patchdb - Use this command to patch an existing database to the service pack level.
  Example:
  ```
  Infordbcl.exe patchdb -databasename:CmdApp -username:sa -password:Sql2012 -
  servername:win2012 -dbtype:App -product:SL -config:test
  ```

infordbcl.exe LinkMultiSite

-databasename: SQL database A name

-username: SQL database A user

-password: SQL database A password

-servername:SQL database A server

-siteid: Database A site id

-sitename: Database A site name

-sitedesc:Database A site description

-sitecurrency:Database A site currency

-sitetype: Database A site type {S,E}

[-sitegroup]: Database A site group

[-timezone]: Database A timezone

-groupname: Multisite group name

-databasename2:SQL database B name

-username2:SQL database B user

-password2: SQL database B password

-servername2: SQL database B server

-siteid2: Database B site id

-sitename2: Database B site name

-sitedesc2: Database B site description

-sitecurrency2: Database B site currency

-sitetype2: Database B site type {S,E}

[-sitegroup2]: Database B site group

[-timezone2]: Database B timezone

[-log]: full path to a log file to redirect to

- linkmultisite - Use this command to link multisite databases
  Example:
  ```
  infordbcl.exe linkmultisite -databasename:"Mongoose1_App" -username:sa -
  password:test123 -servername:win2012 -siteid:1 -sitedesc:"This is site 1" -
  sitetype:S -sitename:"Alpha Site" -groupname:Test -sitecurrency:US -
  databasename2:"Mongoose2_App" -username2:sa -password2:test123 -siteid2:2 -
  sitetype2:S -sitedesc2:"This is site 2" -servername2:"win2012" -
  sitename2:"Beta Site" -sitecurrency2:"US"
  ```

InforDbCL.exe ApplyTrans

-trans: Translation to apply (e.g. "zh-CN", "es-MX") which corresponds to the locale subfolder

-product: Mongoose product (e.g. "Mongoose", "SL") which corresponds to the install subfolder

-config: Mongoose configuration containing the App database and it's referenced Objects database. Only required when the Objects database needs patched.

[-log]: full path to a log file to redirect to

[-alwayson] Establishes a SQL connection to a AlwaysOn Availability Group.

- applytrans - Use this command to apply a translation to a set of databases
  Example:
  ```
  Infordbcl.exe applytrans -config:SL9 -trans:es-mx -product:SL
  ```

InforDbCL.exe ApplyPack

-pack: Industry Pack or Localization to apply (e.g. "automotive", "CN00") which corresponds to the local subfolder

-product: Mongoose product (e.g. "Mongoose", "SL") which corresponds to the install subfolder

-config: Mongoose configuration containing the App database and Forms database information.

[-reportserverurl]: Full URL to report server (if empty reports will not be loaded)

[-reportparentfolder]: The report server parent folder (if empty reports will not be loaded)

[-reportdir]:\tThe local reports folder to upload (defaults to [WinStudioDir]\Report\Reports if not given)

[-log]: Full path to a log file to redirect to

[-alwayson] Establishes a SQL connection to a AlwaysOn Availability Group.

- applypack - Use this command to apply either an Industry Pack or a Localization to a set of databases
  Examples:
  ```
  infordbcl.exe applypack -config:SL9 -product:SL -pack:automotive

  infordbcl.exe applypack -config:SL9 -product:SL -pack:automotive -
  reportserverurl:http://win2012/reportserver -
  reportparentfolder:SyteLineReports
  ```

InforDbCL.exe AddSite

-databasename: SQL database name

-username: SQL server user name

-password: SQL server password

-servername: SQL server name

-product: Mongoose product (e.g. "Mongoose", "SL") which corresponds to the registry key

[-log]: full path to a log file to redirect to

[-siteid]: App database site id

[-sitename]: App database site name

[-sitedesc]: App database site description

[-sitegroup]: App database site group

[-sitecurrency]: App database site currency

[-sitetype]: App database site type {S,E}

[-timezone]: App databse timezone

[-alwayson]: Establishes a SQL connection to a AlwaysOn Availability Group.

[-deletedata]: Deletes all existing data in the existing database.

InforDbCL.exe CopySite

-sourcedatabasename: SQL database name containing existing site to be copied

-sourceusername: SQL server user name

-sourcepassword: SQL server password

-sourceservername: SQL server name containing the source database

-sourcesiteid: App database site id to be copied

-targetdatabasename: SQL database name where the site will be copied

-targetusername: SQL server user name

-targetpassword: SQL server password

-targetservername: SQL server name containing the target database

-product: Mongoose product (e.g. "Mongoose", "SL") which corresponds to the registry key

[-log]: full path to a log file to redirect to

[-alwayson] Establishes a SQL connection to a AlwaysOn Availability Group.

[-analyzeonly] Runs through the copy site process to analyze for potential problems. No copying actually take place in this mode.

[-deletedata] Deletes ALL existing data in the target database. Be careful here!

InforWebCL.exe

This utility contains all configuration tasks associated with IIS and clients. The command must be the first argument. All options in brackets are optional.

Values containing spaces must be in double quotes.The available commands with some examples are:

InforWebCL.exe WebClient:

-product: Mongoose product (e.g. "Mongoose")

[-apppool]: Application pool for web client in IIS ("ASP.NET v4.0" by default)

[-appurl]: Web.config application URL

[-auth]: Authentication type (integer) (0 - builtin, 1 - IIS, 2 - ADFS, 3 - PingFederate)

[-dir]: Virtual directory name for IIS ("WSWebClient" by default)

[-idpfile]: Full path to the idp.properties file; used with PingFederate authentication only

[-log]: Full path to a log file to redirect to

[-path]: Physical path for virtual directory ("wwwroot\WSWebClient" by default)

[-timeout]: Web.config timeout value ("30" by default)

[-remoteido]: Flag for remote IDO {"true", "false"} ("false" by default)

[-remoteidourl]: Web.config URL for config server

[-fedxml]: Web.config federation metadata xml location; used with ADFS authentication only

[-ssl]: Web.config require SSL for cookie authentication, used with PingFederate authentication only

[-stscert]:Web.config federation certificate thumbprint; used with ADFS authentication only

[-trustserver]:Web.config trust server URL; used with ADFS authentication only

- webclient - Use this command to configure the web client
  Examples:
  ```
  Inforwebcl.exe webclient -product:SL
  inforwebcl.exe webclient -product:SL -timeout:20 -remoteido:true -
  remoteidourl:http://server2/idorequestservice/configserver.aspx
  inforwebcl.exe webclient -product:Mongoose -auth:1
  inforwebcl.exe webclient -product:Mongoose -auth:3 -idpfile:idp.properties
  -appurl:http://win2012/wswebclient
  ```

InforWebCL.exe SmartClient:

-appname: Smart Client name, e.g. Mongoose Smart Client

-password: Password for anonymous authentication in IIS

-product: Mongoose product, e.g. "Mongoose"

-user: Useranme for anonymous authentication in IIS

-webserver: Server name

[-configserver]: Server name for remote IDO

[-customcertpassword]: Password for custom Authenticode certificate

[-customcertpath]: Path to a custom Authenticode certificate to sign the Smart Client

[-dir]: Virtual directory name for IIS ("ClientDeploy" by default)

[-log]: full path to a log file to redirect to

[-protocol]: {"http","https"} ("http" by default)

[-remconfig]: Remember configuration group value (integer)

[-remoteido]: Flag for remote IDO {"true", "false"} ("false" by default)

[-root]: ClickOnce physical root folder relative to the WinStudio folder ("ClientDeploy" by default)

[-verson]: Version number to use for the ClickOnce manifest of format x.x.x.x (e.g. 9.2.0.1)

- smartclient - Use this command to configure the smart client
  Examples:
  ```
  inforwebcl.exe smartclient -product:SL -appname:"CloudSuite Smart Client" -
  webserver:win2012 -root:SLClientDeploy -dir:SLClientDeploy -
  user:.\administrator -password:test123

  inforwebcl.exe smartclient -product:Mongoose -appname:"Sweet Mongoose App"
  -webserver:win2012 -user:.\administrator -password:test123
  ```

InforWebCL.exe WebApp

-dir: Virtual directory name for IIS, e.g. IDORequestService

[-apppool]: Application pool in IIS ("ASP.NET v4.0" by default)

[-log]: Full path to a log file to redirect to

[-path]: Full physical path for virtual directory (otherwise "wwwroot\[-dir value]" is used by default)

[-snapshotdelay]: FormAsImage service snapshot delay value (in seconds). Typical value is 10.

[-site]: IS web site ("Default Web Site" by default)

[-timeout]: FormAsImage service timeout value (in seconds). Typical value is 30.

[-webclienturl]: FormAsImage service root web client URL (e.g. http://win2012r2/wswebclient/)

- webapp - Use this command to configure a web application (such as IDORequestService or InboundQueue)
  Examples:
  ```
  inforwebcl.exe webapp -dir:IDORequestService

  inforwebcl.exe webapp -dir:IDORequestService -apppool:DefaultAppPool
  ```

InforWebCL.exe xbap

-appname: Smart Client name, e.g. Mongoose Smart Client

-password: Password for anonymous authentication in IIS

-product: Mongoose product, e.g. "Mongoose", which corresponds to the install subfolder

-user: Useranme for anonymous authentication in IIS

-webserver: Server name

[-configserver]: Server name for remote IDO

[-customcertpassword]: Password for custom Authenticode certificate

[-customcertpath]: Path to a custom Authenticode certificate to sign the Smart Client

[-dir]: Virtual directory name for IIS ("XBAP" by default)

[-log]: full path to a log file to redirect to

[-protocol]: {"http","https"} ("http" by default)

[-remconfig]: Remember configuration group value (integer)

[-remoteido]: Flag for remote IDO {"true", "false"} ("false" by default)

[-root]: ClickOnce physical root folder ("XBAPClientDeploy" by default)

[-verson]: Version number to use for the ClickOnce manifest of format x.x.x.x (e.g. 9.2.0.1)

- xbap - Use this command to configure the smart client via web (aka XBAP client)
  Examples: configure smart client via web with default values
  ```
  inforwebcl.exe xbap -appname:"Smart Client via Browser" -product:Mongoose -
  webserver:win2012 -user:win2012\administrator -password:test123

  inforwebcl.exe xbap -appname:"CloudSuite via Browser" -product:SL -
  webserver:win2012 -user:win2012\administrator -password:test123
  ```

InforWebCL.exe EndUser

-configserver: Full config server URL

-product: Mongoose product, e.g. "Mongoose"

-shortcutname: Name for the shortcut

[-description]:Description for the shortcut

[-icon]: Icon for the shortcut ({InforDir}\{Product}\ERP.ico by default)

[-log]: full path to a log file to redirect to

- enduser - Use this command to configure the end user client shortcut
  Example: create a desktop shortcut to Mongoose
  ```
  inforwebcl.exe enduser -configserver:http://win2012/idorequestservice/
  configserver.aspx -product:Mongoose -shortcutname:"Infor Mongoose"
  ```

InforWebCL.exe PingFederate

-appname: PingFederate will use this name for your application

-appdesc: Short description of your application

-entityurl: URL to the web client; requires https and port. i.e. - https://myserver:44300/WSWebClient

-outputpath: Local directory where the sp.properties file will be written

[-log]: Full path to a log file to redirect to

- PingFederate
  Example:
  ```
  inforwebcl.exe pingfederate -appname:MyApp -appdesc:"A great app" -
  entityurl:http://win2012/wswebclient -
  outputpath:c:\users\administrators\desktop
  ```

InforServiceCL.exe

This utility contains the configuration tasks dealing with Windows services and replication.

Replication

The available commands with some examples are:

-server: Name of server to configure

-repuser: Username to give permission of queue to and set Replication service logon (LocalSystem by default)

-reppassword: Password of user to give permission of queue to and set Replication service logon

-listeneruser: Username to give permission of queue to and set ReplQListener service logon (LocalSystem by default)

-listenerpassword: Password of user to give permission of queue to and set ReplQListener service logon

[-log]: full path to a log file to redirect to
Example:

```
inforservicecl.exe replication -server:win2012 -repuser:win2012\administrator
-reppassword:test123 -listeneruser:win2012\administrator -
listenerpassword:test123
```

setserviceaccount

-LogonAcct: Security account under which the service will be run (e.g. Domain\User, LocalSystem, NetworkService)

-Service: Windows service to set the credentials on (if TaskMan command was used this is set to 'TaskMan')

[-LogonPwd]: Password for the security account

[-Server]: Name of the server to configure service on. If not specified, the local machine will be used

[-SkipValidation]: Indicate if security account and password validation can be skipped. Default value is 'false'

[-ServerAcct]: Used to specify whether this configuration task should be performed by an account other than the currently logged on user. Generally, only needed if being done on a remote server.

[-ServerPwd]: Password for server account

- Examples:
  ```
  inforservicecl.exe setserviceaccount -service:TaskMan -
  logonacct:networkservice

  inforservicecl.exe setserviceaccount -service:TaskMan -
  logonacct:win2012\administrator -logonpwd:test123
  ```

Report Services

Use this command to Configure Report Services and deploy reports to the Report Server.

-URL: Required) Report Server URL (e.g. http://win2012r2/reportserver)

[-folder]: Report Server Parent Folder (e.g. MongooseReports). If none is given a '{Product}Reports' string will be generated

[-dir]: Full path to the reports directory. If none given, the Report\Reports folder under the WinStudio directory will be used

[-reportsonly]: Import SSRS Reports (RDLs) only; no IDO based reporting setup

Examples:

```
inforservicecl.exe reportservices -url:http://report.server.com/ReportServer -
folder:SyteLineReport -dir:"C:\Program Files (x86)\Infor\SyteLine\Reports\"
```

```
inforservicecl.exe reportservices -url:http://win2012r2/ReportServer -
reportsonly
```

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